AI in Data Analysis: Empowering PruBSN with Copilot in Excel

From Raw Data to Insights in Minutes





Dr. Dwi Pebrianti

Module 3

Copilot is your analytical assistant, not a replacement for your expertise

MEET THE TRAINER



DR DWI PEBRIANTI

Center for Unmanned Technology (CUTe)

Education

PhD. (Artificial System Science)
Chiba University, Japan
M.Eng (Precision Engineering)
The University of Tokyo, Japan
BEng. (Control), Universitas Indonesia

Area of Expertise

Robotics, Unmanned Aerial Vehicle,
Machine Learning and Artificial Intelligence

Training Experience

Autonomous Robot
Intelligent Control
Optimization for Engineering
Data science
AI for Business

Module Covered

Module 3:
AI in Data Analysis

Module 4:
AI for Productivity

KEEP THE MOMENTUM

Participants Preparation

01	LAPTOP	Microsoft 365 Copilot licence, Excel and Teams
02	RAW DATA	**PRUBSN_Tier 1_Business Data.xlsx** https://github.com/uychan/PruBSN
03	SANDBOX CHANNEL	Safe workflow testing

Microsoft Copilot

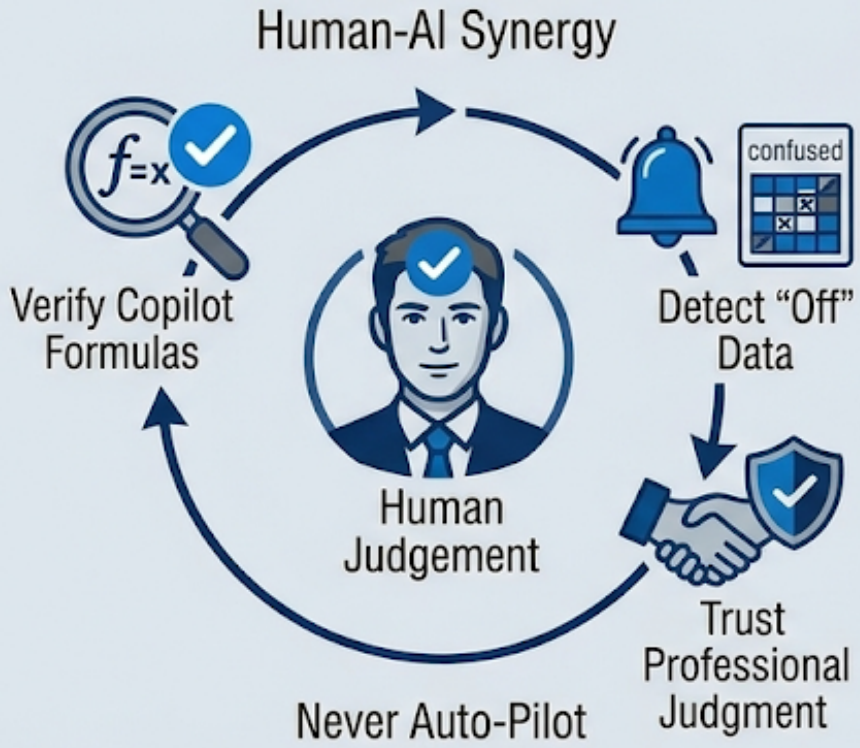
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Data Analysis Philosophy: The Copilot Paradigm Shift

The "80/20" Rule: The Historical Shift



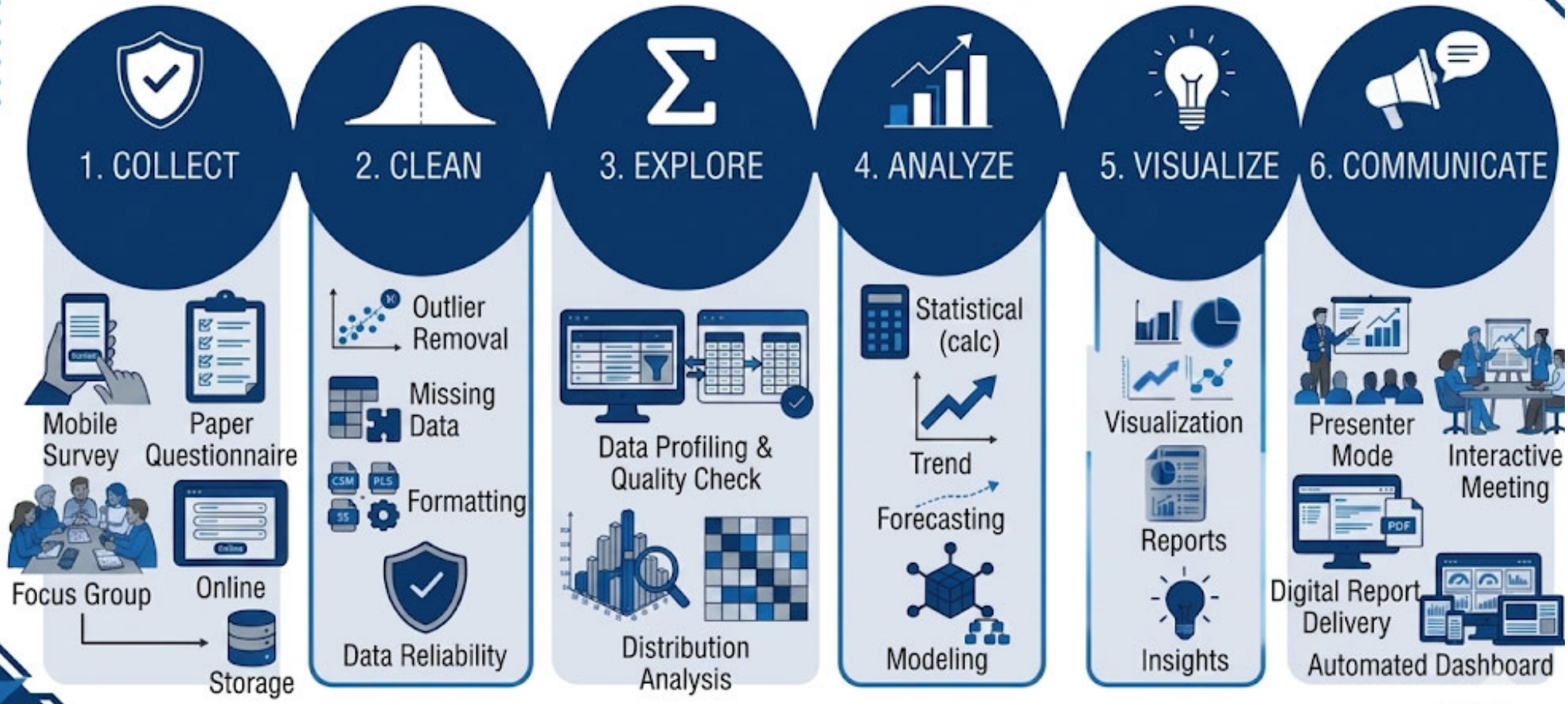
Human-in-the-Loop: Trust and Validate



Copilot empowers your expertise. You are the critical, human element for strategic insights.

DATA CYCLE: AN INTEGRATED GUIDE

Foundation **Transformation** **Impact**





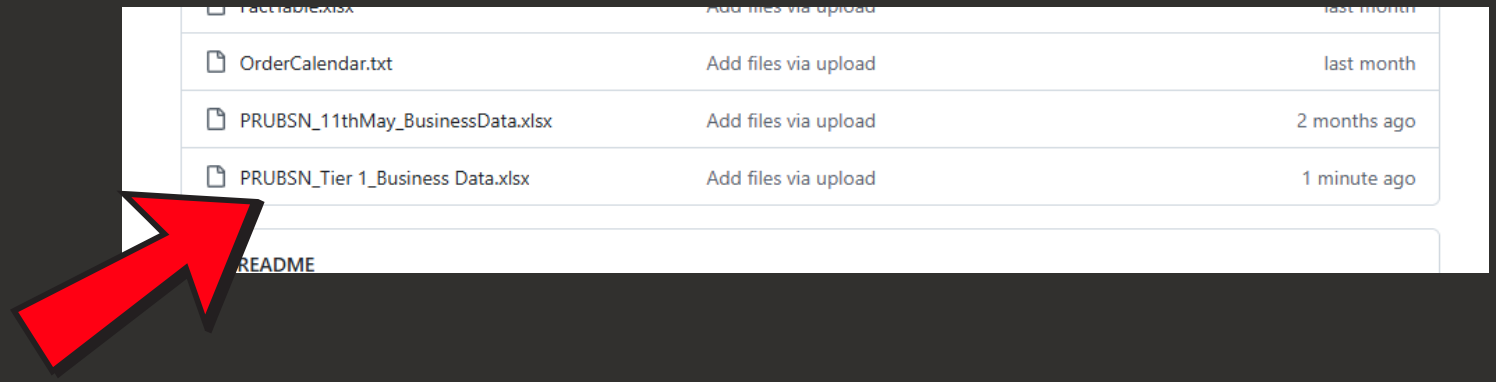
HANDS ON

Step 1: Data Collection



Link to sample file

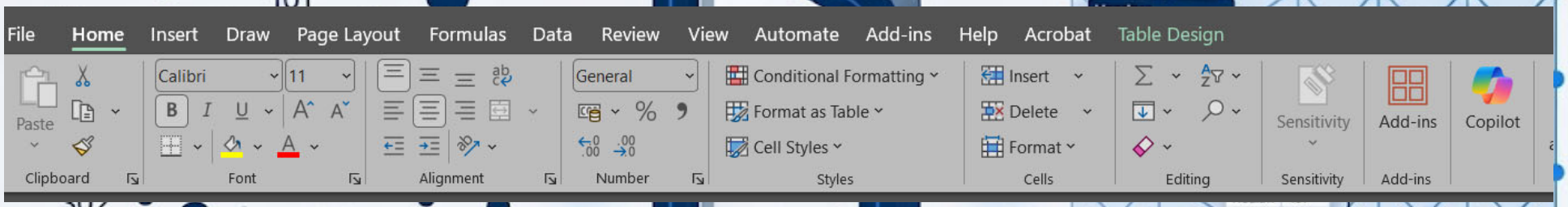
https://github.com/uychan/PruBSN

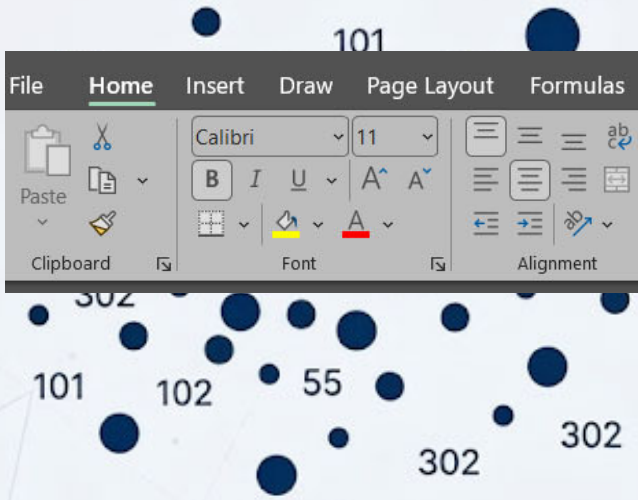


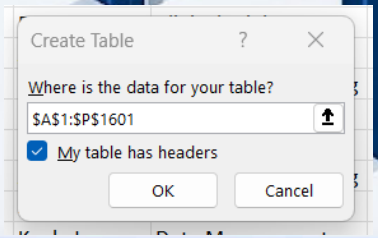
Ctrl + T is the essential ignition key for your AI assistant.

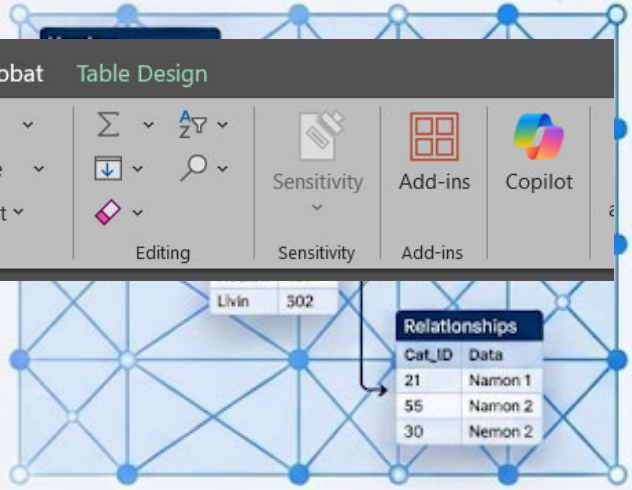
Unreadable by AI

Headers, data types, and relationships unlocked.

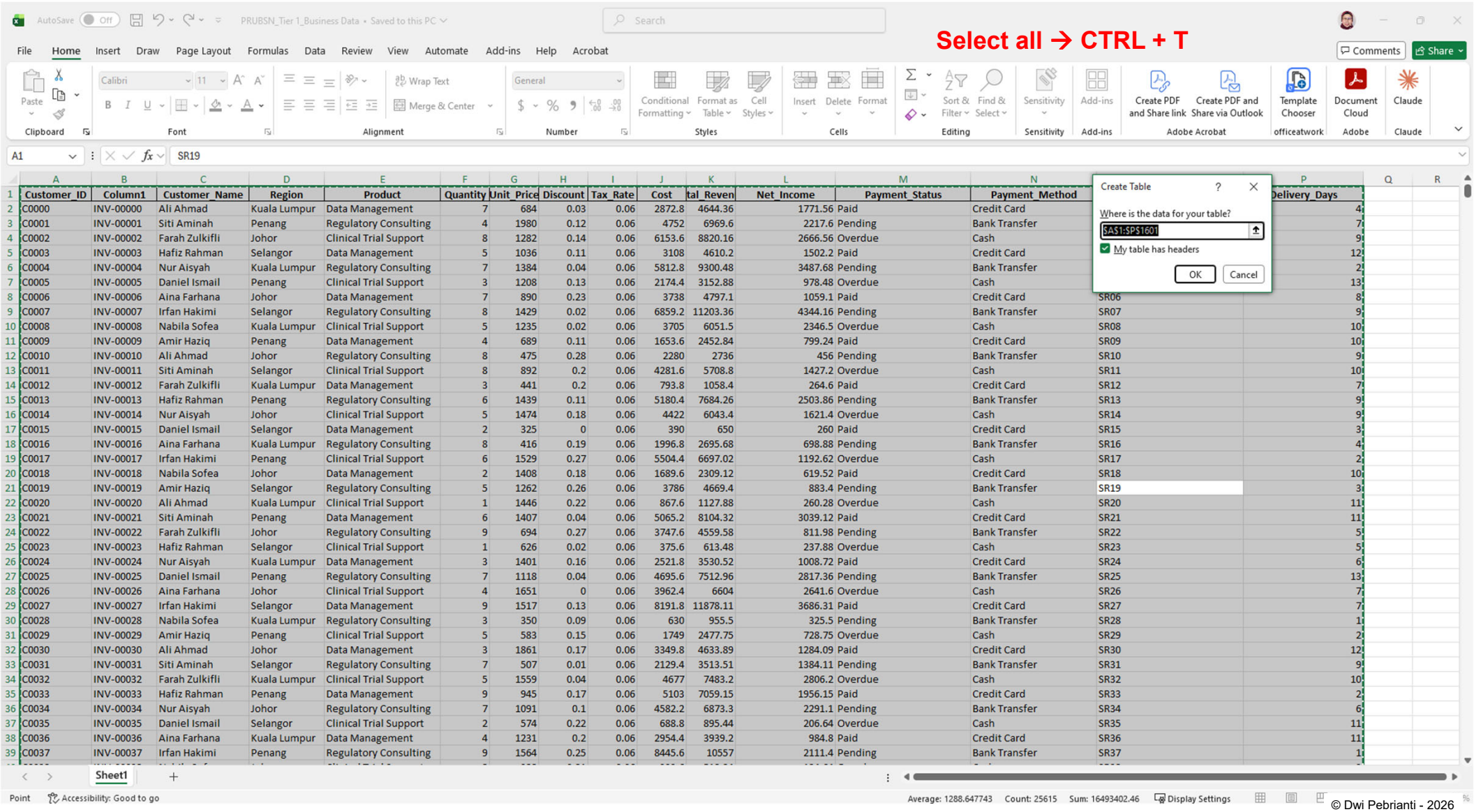






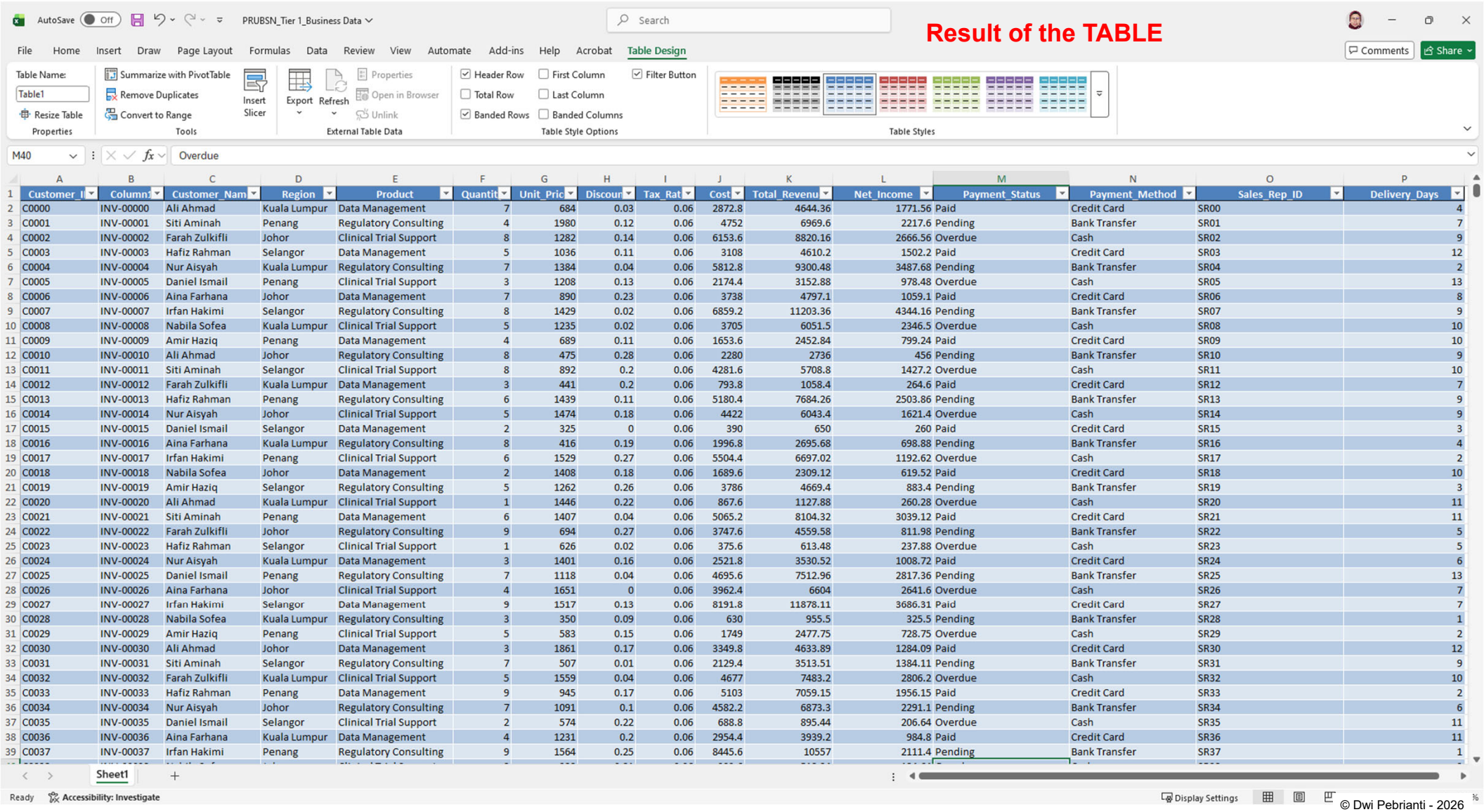


Why? Copilot needs a structured Table to understand the context of your data. It cannot 'read' a loose range of cells.



Customer_ID	Column1	Customer_Name	Region	Product	Quantity	Unit_Price	Discount	Tax_Rate	Cost	tal_Reven	Net_Income	Payment_Status	Payment_Method		Delivery_Days
C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36	1771.56	Paid	Credit Card		4
C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6	2217.6	Pending	Bank Transfer		7
C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16	2666.56	Overdue	Cash		9
C0003	INV-00003	Hafiz Rahman	Selangor	Data Management	5	1036	0.11	0.06	3108	4610.2	1502.2	Paid	Credit Card		12
C0004	INV-00004	Nur Aisyah	Kuala Lumpur	Regulatory Consulting	7	1384	0.04	0.06	5812.8	9300.48	3487.68	Pending	Bank Transfer		2
C0005	INV-00005	Daniel Ismail	Penang	Clinical Trial Support	3	1208	0.13	0.06	2174.4	3152.88	978.48	Overdue	Cash		13
C0006	INV-00006	Aina Farhana	Johor	Data Management	7	890	0.23	0.06	3738	4797.1	1059.1	Paid	Credit Card	SR06	8
C0007	INV-00007	Irfan Hakimi	Selangor	Regulatory Consulting	8	1429	0.02	0.06	6859.2	11203.36	4344.16	Pending	Bank Transfer	SR07	9
C0008	INV-00008	Nabila Sofea	Kuala Lumpur	Clinical Trial Support	5	1235	0.02	0.06	3705	6051.5	2346.5	Overdue	Cash	SR08	10
C0009	INV-00009	Amir Haziq	Penang	Data Management	4	689	0.11	0.06	1653.6	2452.84	799.24	Paid	Credit Card	SR09	10
C0010	INV-00010	Ali Ahmad	Johor	Regulatory Consulting	8	475	0.28	0.06	2280	2736	456	Pending	Bank Transfer	SR10	9
C0011	INV-00011	Siti Aminah	Selangor	Clinical Trial Support	8	892	0.2	0.06	4281.6	5708.8	1427.2	Overdue	Cash	SR11	10
C0012	INV-00012	Farah Zulkifli	Kuala Lumpur	Data Management	3	441	0.2	0.06	793.8	1058.4	264.6	Paid	Credit Card	SR12	7
C0013	INV-00013	Hafiz Rahman	Penang	Regulatory Consulting	6	1439	0.11	0.06	5180.4	7684.26	2503.86	Pending	Bank Transfer	SR13	9
C0014	INV-00014	Nur Aisyah	Johor	Clinical Trial Support	5	1474	0.18	0.06	4422	6043.4	1621.4	Overdue	Cash	SR14	9
C0015	INV-00015	Daniel Ismail	Selangor	Data Management	2	325	0	0.06	390	650	260	Paid	Credit Card	SR15	3
C0016	INV-00016	Aina Farhana	Kuala Lumpur	Regulatory Consulting	8	416	0.19	0.06	1996.8	2695.68	698.88	Pending	Bank Transfer	SR16	4
C0017	INV-00017	Irfan Hakimi	Penang	Clinical Trial Support	6	1529	0.27	0.06	5504.4	6697.02	1192.62	Overdue	Cash	SR17	2
C0018	INV-00018	Nabila Sofea	Johor	Data Management	2	1408	0.18	0.06	1689.6	2309.12	619.52	Paid	Credit Card	SR18	10
C0019	INV-00019	Amir Haziq	Selangor	Regulatory Consulting	5	1262	0.26	0.06	3786	4669.4	883.4	Pending	Bank Transfer	SR19	3
C0020	INV-00020	Ali Ahmad	Kuala Lumpur	Clinical Trial Support	1	1446	0.22	0.06	867.6	1127.88	260.28	Overdue	Cash	SR20	11
C0021	INV-00021	Siti Aminah	Penang	Data Management	6	1407	0.04	0.06	5065.2	8104.32	3039.12	Paid	Credit Card	SR21	11
C0022	INV-00022	Farah Zulkifli	Johor	Regulatory Consulting	9	694	0.27	0.06	3747.6	4559.58	811.98	Pending	Bank Transfer	SR22	5
C0023	INV-00023	Hafiz Rahman	Selangor	Clinical Trial Support	1	626	0.02	0.06	375.6	613.48	237.88	Overdue	Cash	SR23	5
C0024	INV-00024	Nur Aisyah	Kuala Lumpur	Data Management	3	1401	0.16	0.06	2521.8	3530.52	1008.72	Paid	Credit Card	SR24	6
C0025	INV-00025	Daniel Ismail	Penang	Regulatory Consulting	7	1118	0.04	0.06	4695.6	7512.96	2817.36	Pending	Bank Transfer	SR25	13
C0026	INV-00026	Aina Farhana	Johor	Clinical Trial Support	4	1651	0	0.06	3962.4	6604	2641.6	Overdue	Cash	SR26	7
C0027	INV-00027	Irfan Hakimi	Selangor	Data Management	9	1517	0.13	0.06	8191.8	11878.11	3686.31	Paid	Credit Card	SR27	7
C0028	INV-00028	Nabila Sofea	Kuala Lumpur	Regulatory Consulting	3	350	0.09	0.06	630	955.5	325.5	Pending	Bank Transfer	SR28	1
C0029	INV-00029	Amir Haziq	Penang	Clinical Trial Support	5	583	0.15	0.06	1749	2477.75	728.75	Overdue	Cash	SR29	2
C0030	INV-00030	Ali Ahmad	Johor	Data Management	3	1861	0.17	0.06	3349.8	4633.89	1284.09	Paid	Credit Card	SR30	12
C0031	INV-00031	Siti Aminah	Selangor	Regulatory Consulting	7	507	0.01	0.06	2129.4	3513.51	1384.11	Pending	Bank Transfer	SR31	9
C0032	INV-00032	Farah Zulkifli	Kuala Lumpur	Clinical Trial Support	5	1559	0.04	0.06	4677	7483.2	2806.2	Overdue	Cash	SR32	10
C0033	INV-00033	Hafiz Rahman	Penang	Data Management	9	945	0.17	0.06	5103	7059.15	1956.15	Paid	Credit Card	SR33	2
C0034	INV-00034	Nur Aisyah	Johor	Regulatory Consulting	7	1091	0.1	0.06	4582.2	6873.3	2291.1	Pending	Bank Transfer	SR34	6
C0035	INV-00035	Daniel Ismail	Selangor	Clinical Trial Support	2	574	0.22	0.06	688.8	895.44	206.64	Overdue	Cash	SR35	11
C0036	INV-00036	Aina Farhana	Kuala Lumpur	Data Management	4	1231	0.2	0.06	2954.4	3939.2	984.8	Paid	Credit Card	SR36	11
C0037	INV-00037	Irfan Hakimi	Penang	Regulatory Consulting	9	1564	0.25	0.06	8445.6	10557	2111.4	Pending	Bank Transfer	SR37	1

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Customer_I	Columns	Customer_Nam	Region	Product	Quantit	Unit_Pric	Discoun	Tax_Rat	Cost	Total_Revenu	Net_Income	Payment_Status	Payment_Method	Sales_Rep_ID	Delivery_Days
C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7	684	0.03	0.06	2872.8	4644.36	1771.56	Paid	Credit Card	SR00	4
C0001	INV-00001	Siti Aminah	Penang	Regulatory Consulting	4	1980	0.12	0.06	4752	6969.6	2217.6	Pending	Bank Transfer	SR01	7
C0002	INV-00002	Farah Zulkifli	Johor	Clinical Trial Support	8	1282	0.14	0.06	6153.6	8820.16	2666.56	Overdue	Cash	SR02	9
C0003	INV-00003	Hafiz Rahman	Selangor	Data Management	5	1036	0.11	0.06	3108	4610.2	1502.2	Paid	Credit Card	SR03	12
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C0031	INV-00031	Siti Aminah	Selangor	Regulatory Consulting	7	507	0.01	0.06	2129.4	3513.51	1384.11	Pending	Bank Transfer	SR31	9
C0032	INV-00032	Farah Zulkifli	Kuala Lumpur	Clinical Trial Support	5	1559	0.04	0.06	4677	7483.2	2806.2	Overdue	Cash	SR32	10
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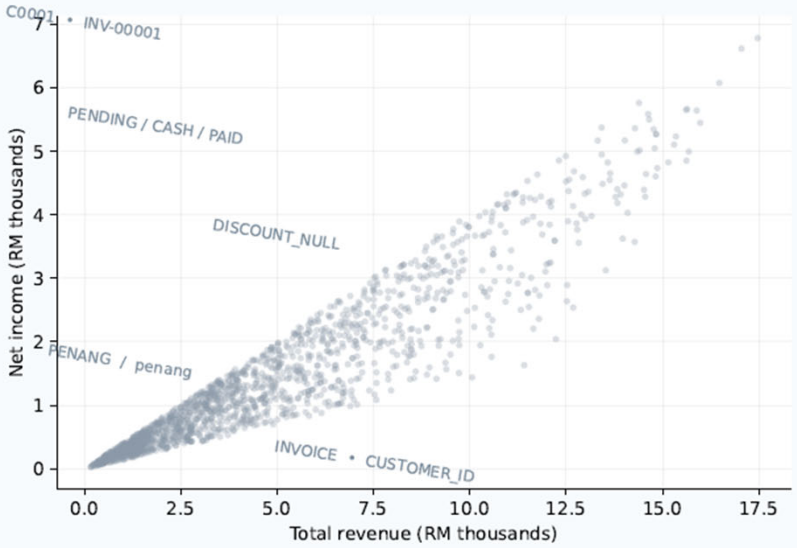
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FROM RAW BUSINESS DATA TO ACTIONABLE INSIGHT

PRU BSN Tier 1 Business Data | Exploratory Data Analysis | 1,600 transactions

01 THE RAW SCENE

Unstructured view of 1,600 records

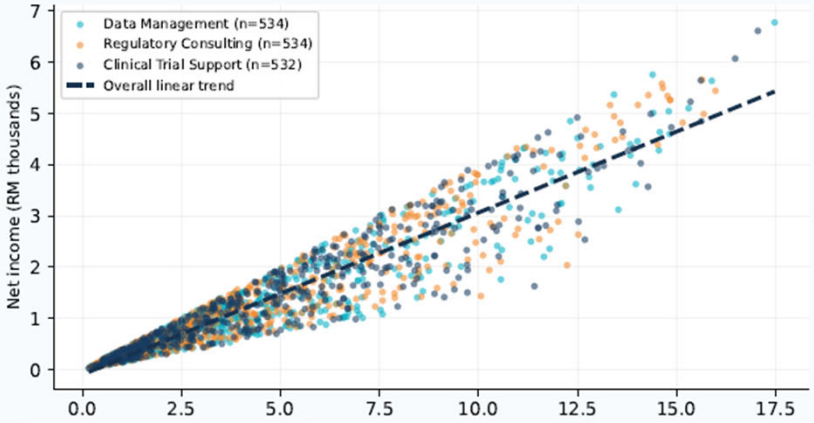


DATA QUALITY SIGNALS

- Missing Discount values: 1
- Region spelling inconsistency: Penang / penang
- 16 columns across 1,600 transactions

02 THE EXTRACTED INSIGHT

REVENUE–NET INCOME CORRELATION r = 0.943



TOTAL REVENUE	TOTAL NET INCOME	PAID TRANSACTIONS
RM 7.35M	**RM 2.17M**	**534**
All transactions	Margin 29.5%	of 1,600 records

PATTERN IDENTIFIED:

Strong positive revenue–profit relationship; product groups overlap.

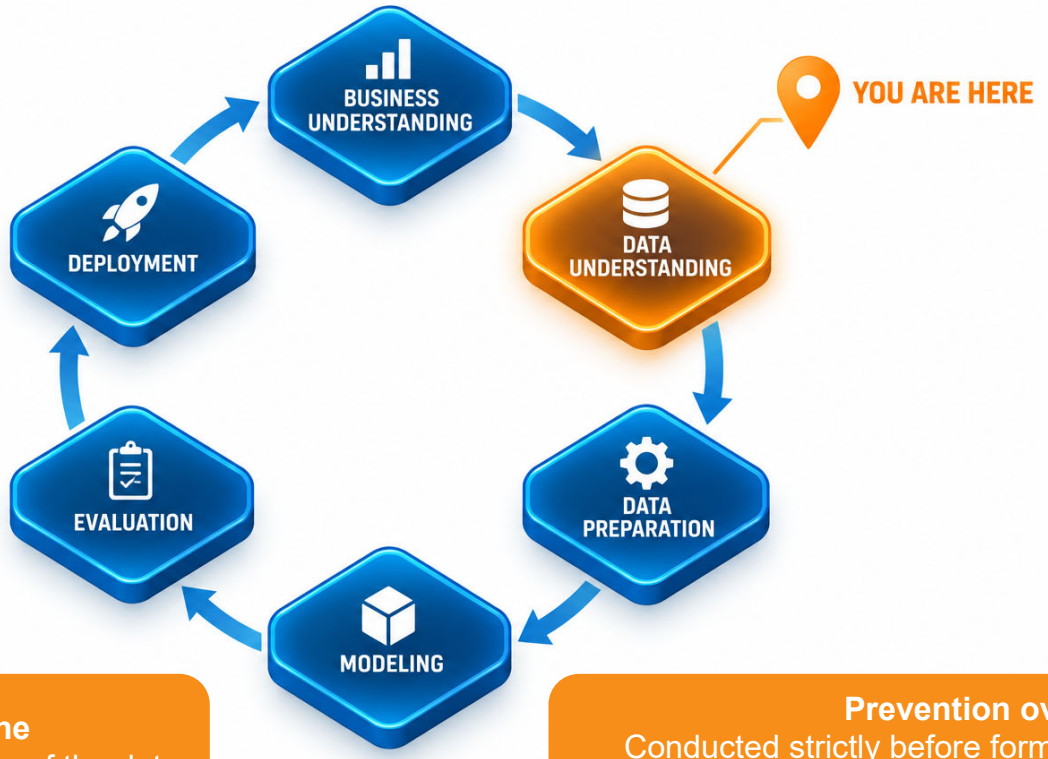
EDA reveals data-quality issues, quantifies business performance, and identifies relationships before predictive modelling.

Case study in Insurance : claim approval, fraud detection, customer segmentation, personalized insurance

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CRISP – DM : The Industry Standard Pipeline

Cross-Industry Standard Process for Data Mining



The industry standard pipeline
EDA is not an optional phase. It is the core of the data understanding phase in the CRISP-DM pipeline

Prevention over cure
Conducted strictly before formal hypothesis testing or predictive modeling. Bypassing EDA risks building advanced algorithms on flawed assumption. Garbage in, garbage out

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Why Data Exploration Matters

Process of **examining** data to understand its structure, quality, and contents before using it for **analysis or visualization**.



- **Ensures** data **cleanliness and accuracy**
- **Helps identify** incomplete or inconsistent records
- **Saves time** by catching **errors** before building visuals



HANDS ON

Step 2: Data Cleaning



Data Cleaning Essentials

What is Data Cleaning?

Detecting and correcting inaccurate, incomplete, or inconsistent data in a dataset

Why Clean Data?

To improve data quality — accurate, reliable, and ready for analysis

Key Data Cleaning Steps

- Remove errors, blanks, or duplicates
- Replace incorrect values
- Change or fix data types
- Standardize text formats and column names
- Handle missing or null data



THE DATA ANATOMY MATRIX

NUMERICAL (Continuous & Discrete)	CATEGORICAL (Nominal)	ORDINAL	TIME SERIES
Definition: Measured or counted values.	**Definition:** Unordered labels or groups.	**Definition:** Categorical data with a defined sequence.	**Definition:** Data collected sequentially over time.
Example: Premium amount, Claim amount, Customer age, Number of claims.	**Example:** Policy type, Claim type, Payment method, Branch region.	**Example:** Risk level (Low-Medium-High), Claim priority, Customer satisfaction rating.	**Example:** Daily claim counts, Monthly premium revenue, Claim trends by month.
Tools: Mean, Median, Standard Deviation.	**Tools:** Frequency counts, Proportions, Mode.	**Tools:** Median, Percentiles, Rank correlation.	**Tools:** Trend analysis, Seasonality checks.

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HANDS ON

Common errors





movie_title	num_critic_for_reviews	duration	DIRECTOR_facebook_likes	actor_3_facebook_likes	ACTOR_1_facebook_likes	gross
Avatar?ÿ	723	178	10	855	1000	760505847
Pirates of the Caribbean: At World's End?ÿ	302		563	1000	40000	309404152
Spectre?ÿ	602	148	20	161	11000	200074175
The Dark Knight Rises?ÿ	813		22000	23000	27000	448130642
John Carter?ÿ	462	132	"475"	530	640	73058679
Spider-Man 3?ÿ	392	156	23	4000	24000	336530303
Tangled?ÿ	324		15	284	799	200807262
Avengers: Age of Ultron?ÿ	635	141	10	19000	26000	458991599
Avengers: Age of Ultron?ÿ	635	141	10	19000	26000	458991599
Harry Potter and the Half-Blood Prince?ÿ	375	153	282	10000	25000	301956980
Batman v Superman: Dawn of Justice?ÿ	673	183		2000	15000	330249062
Superman Returns?ÿ	434	169		903	18000	200069408
Quantum of Solace?ÿ	403	106	395	393	451	168368427

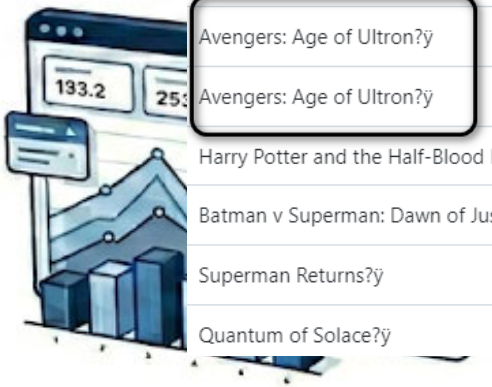
Noisy data

Incorrect formatting

Inconsistent headers

Duplication

Missing data



Ask co-pilot:

For data cleaning, analyze the data:

What is wrong with it?

Are there any errors?

- Into Copilot
- Give the excel file
- analyze this data the summarize What is wrong with it? Are there any errors?

Highlight :
C0100
C0200
C0300
C0400



tips:

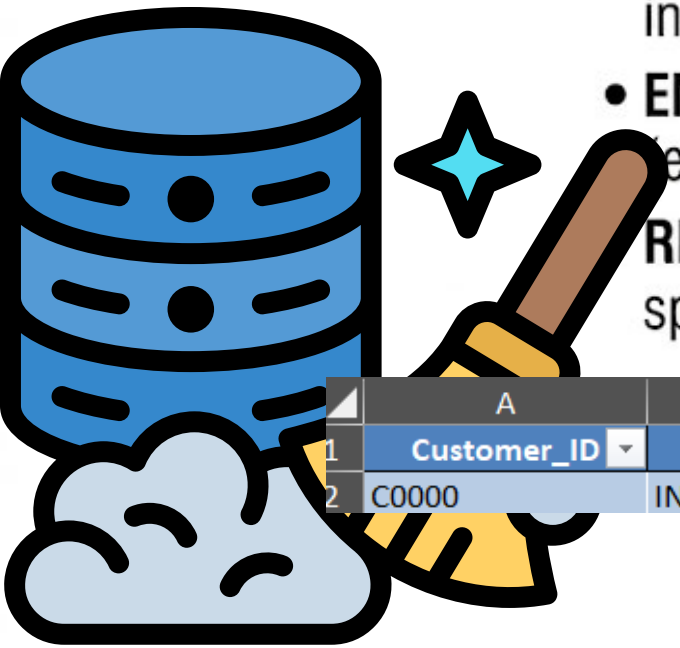
AutoFit Column Width,
Press:Alt→H→0→1

AutoFit Row Height,
Press:Alt→HOA

LET'S FIX THE ERROR:
Challenge 1 - Column Renaming.

Why is this Important? (Benefits)

- **IMPROVES CLARITY AND UNDERSTANDING:** Use meaningful names like CustomerName or SalesAmount instead of vague ones like Col1 or Value.
- **ENSURE CONSISTENCY:** Standardize data formats (e.g., STD_ID) across all tables and databases.
- **REFLECTS TRUE MEANING:** Change generic terms to specific ones, like Vague name... to Total_Sales.



	A	B	C	D	E	F
1	Customer_ID	Column1	Customer_Name	Region	Product	Quantity
2	C0000	INV-00000	Ali Ahmad	Kuala Lumpur	Data Management	7

LET'S FIX THE ERROR:
Challenge 1 - Column Renaming.

Co-pilot: Find column headers with generic labels and suggest appropriate labels for them.

Inappropriate Column Name	Suggested Appropriate Name
Col1	Customer_ID
A	Order_Date
Data_2023	Sales_Amount





LET'S FIX THE ERROR:

Challenge 2 - Duplicate Data.

Identical or repeated records Identical or repeated records appearing more than once in the dataset.



Causes of Duplicate Data:

Main Causes	Implications (Summary)
(i) Repeated data imports or merges	• Identical records appear more than once.
(ii) Manual entry errors	• Leading to distorted results in analysis and reporting.
(iii) Multiple sources collecting the same information	• Causes discredit and poor decision making.
(iv) System glitches or unsynchronized databases	• Increases storage costs.



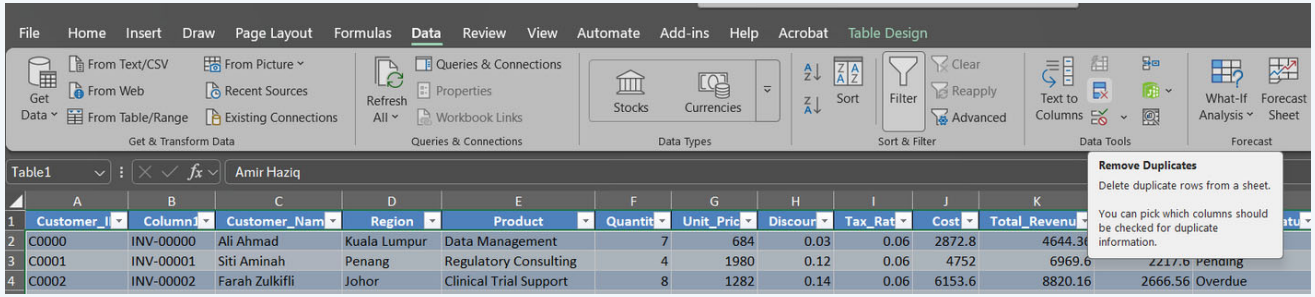
Prompt: Find duplicate data

✅ How to handle this in Excel (quick steps)

If you want to remove duplicates directly in Excel:

1. Select the **entire table**
2. Go to **Data → Remove Duplicates**
3. Select **all columns**
4. Click **OK**

Excel will remove exactly **one duplicate row** (the `C0100 / INV-00100` entry).





Data Cycle Guide: Addressing Missing Values (Clean)

2. CLEAN
Address structure & errors

Challenge 3: Missing Values

Data gaps where information is blank, null, or unreadable
(e.g., N/A, empty cells)

REASON ANALYSIS
(Causes of Data Gaps)

1. **Human error**
   Incorrect data entry, typos, misreading info
2. **System or sensor failure**
   Sensor malfunctions, power loss, database connectivity issues
3. **Conditional or intentional skipping**
   Skip logic in surveys, deliberate non-response
4. **Privacy or anonymity constraints**
   Data masking, removal for compliance & privacy

REMEDIATION STRATEGIES
(Techniques for Imputation & Handling)

1. **Remove rows** – if few or irrelevant
2. **Replace values** – with mean, median, or "Unknown"
3. **Fill up / down** – for sequential data
4. **Flag missing** – keep track for later analysis

Why? Copilot needs structure. Imputation or flagging provides **necessary structure.**



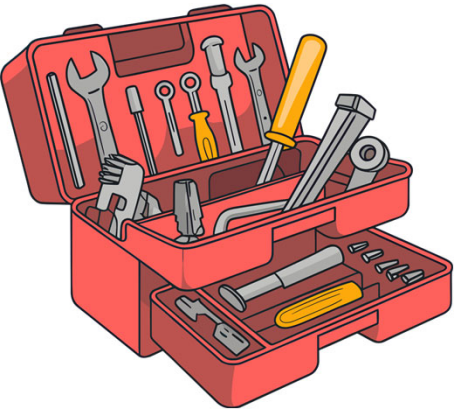
This is *your* expertise!

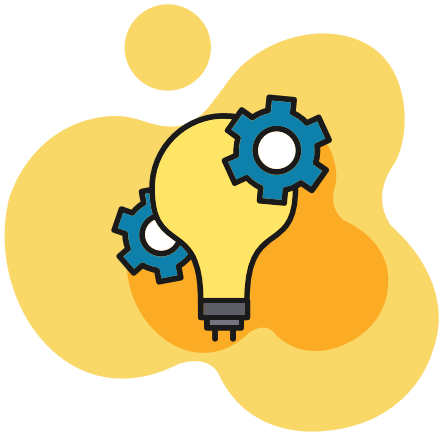
NotebookLM

I have analyzed your data file and can confirm that **Row 402** (which corresponds to Customer ID **C0400**, Invoice **INV-00400**) is indeed missing a value in the **Discount** column.

Verification details for Row 402:

- **Product:** Regulatory Consulting
- **Quantity:** 1
- **Unit Price:** 983
- **Total Revenue:** 707.76
- **Discount:** *Missing (NaN)*





Suggested Best Way to Fill the Value

In your dataset, the values follow a specific mathematical relationship. Instead of using a simple average (mean) or median, the most accurate way is to **reverse-calculate** the discount using the existing financial data in that row.

Based on the other rows in your file, the formula used is:

$$\text{Total Revenue} = (\text{Quantity} \times \text{Unit Price}) \times (1 - \text{Discount})$$

The Calculation:

1. **Gross Amount:** $1 \times 983 = 983$
2. **Revenue Ratio:** $707.76 / 983 = 0.72$
3. **Discount Value:** $1 - 0.72 = 0.28$

Recommended Value: You should enter **0.28** (or 28%) into the Discount cell for Row 402. This perfectly aligns with the Total Revenue and Net Income already recorded for that entry.

How to apply this in Excel/CSV:

1. Locate cell **H402** (assuming Column H is Discount).
2. Type `0.28`.
3. If you want to automate this for any future missing values, you can use this Excel formula in the Discount column:
   `=1 - (Total_Revenue / (Quantity * Unit_Price))`
   =1-([@[Total_Revenue]]/([@Quantity]*[@[Unit_Price]]))



HANDS ON

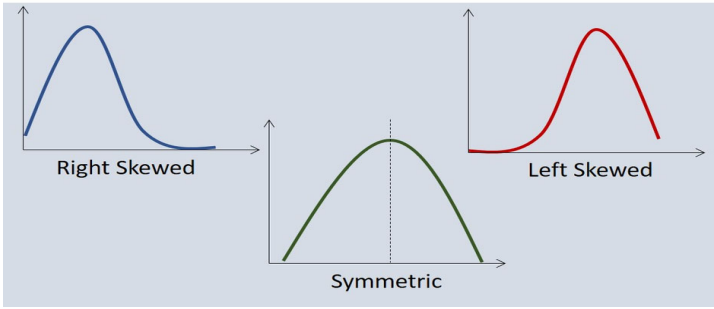
Step 3: Data Exploration

Please perform a distribution analysis.

Share the outputs.

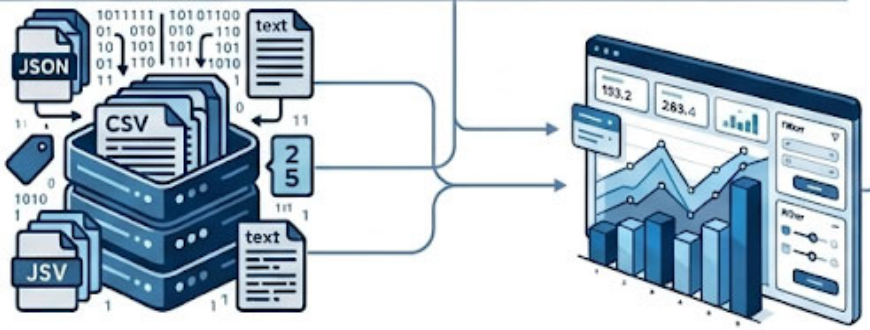


Outcomes:



e.g., Distribution analysis

Field Type	Distribution Shape
Regions	Moderately imbalanced
Products	Balanced (3 categories)
Payment Status	Balanced (rule-based)
Quantity	Right-skewed
Prices & Revenue	Heavily right-skewed
Tax Rate	Narrow, stable
Delivery Days	Compact, uniform



Overall Data Quality Assessment

✅ Strengths
- Rich transactional detail
- High completeness
- Logical relationships preserved
- Suitable for customer, revenue, and operational analysis

❌ Weaknesses
- Numeric formatting issues
- Scientific notation artifacts
- Outliers without flags
- Minor text inconsistencies

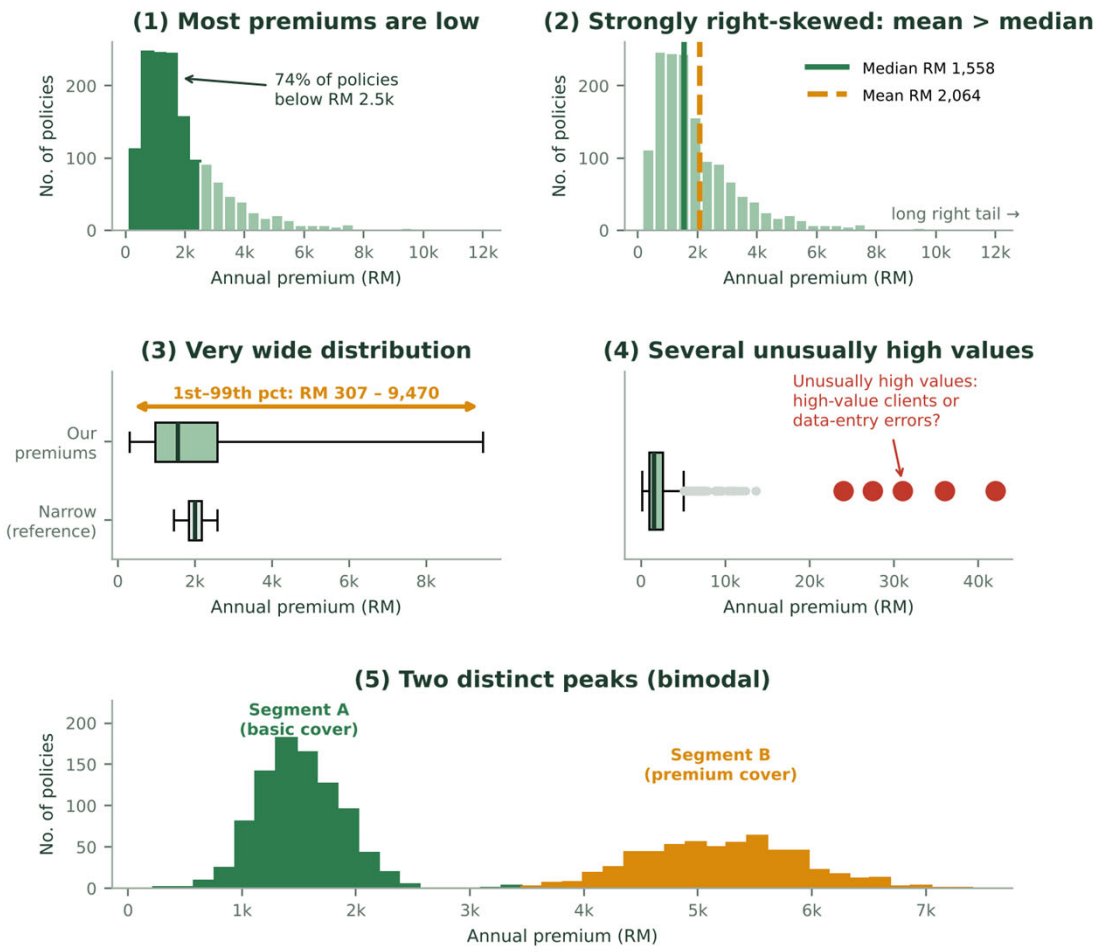
Recommended Next Steps

- **Normalize numeric formats**: Convert scientific notation to standard decimals, round monetary fields consistently.
- **Validate outliers**: Apply IQR/Z-score checks on Quantity and Revenue, flag extreme invoices for review.
- **Clean categorical text**: Standardize capitalizatio trim and correct corrupted strings.
- **Recompute financials**: Recalculate Cost, Tax, Revenue, Net_Income from base fields, compare against stored values.
- **Prepare for analytics**: Create derived metrics (ARPU, overdue rate, avg delivery time), segment customers by region/service.



Case Study – Insurance

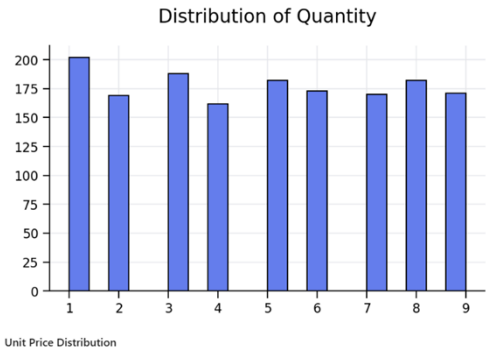
Observation	Possible business interpretation
Most premiums are low (1)	Lower-premium policies may be more common among customers.
Distribution is strongly right-skewed (2)	A small number of high-premium customers may increase the mean.
Very wide distribution (3)	Customers have diverse premium amounts, potentially requiring different products.
Several unusually high values (4)	Investigate whether they are high-value customers or data-entry errors.
Two distinct peaks (5)	There may be different customer segments worth analyzing.

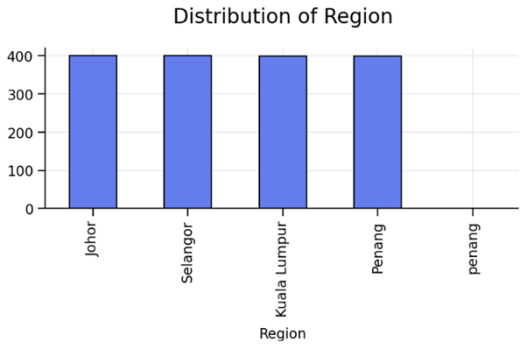


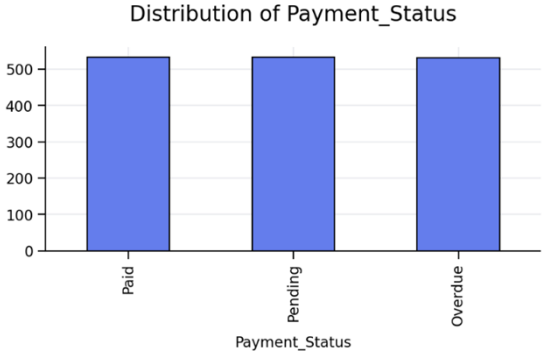
Illustrative simulated premium data

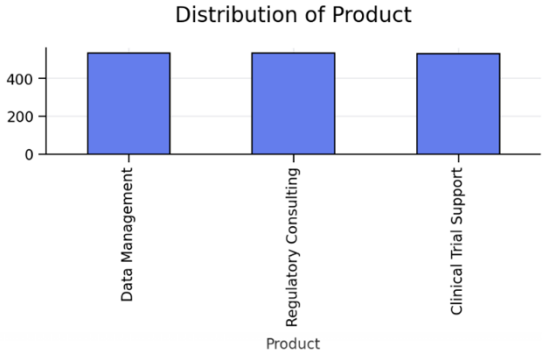
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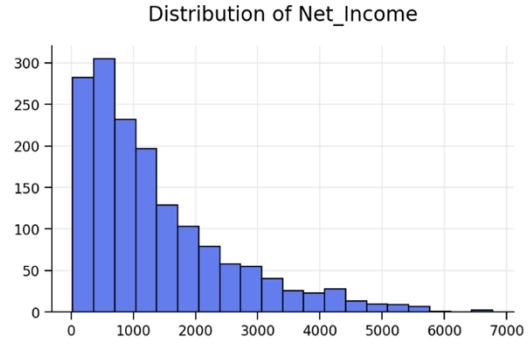
for each of the feature, generate the graph of the distribution

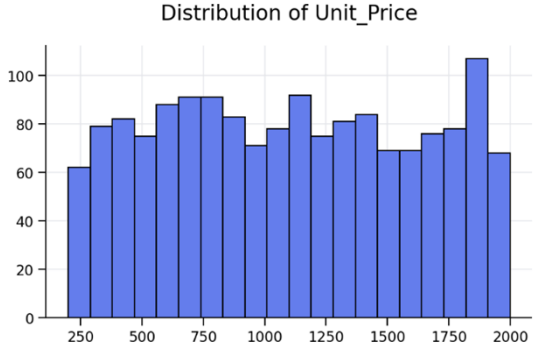












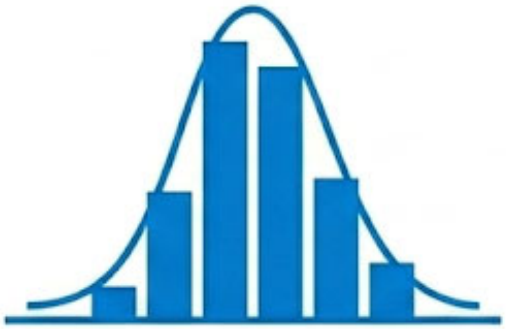
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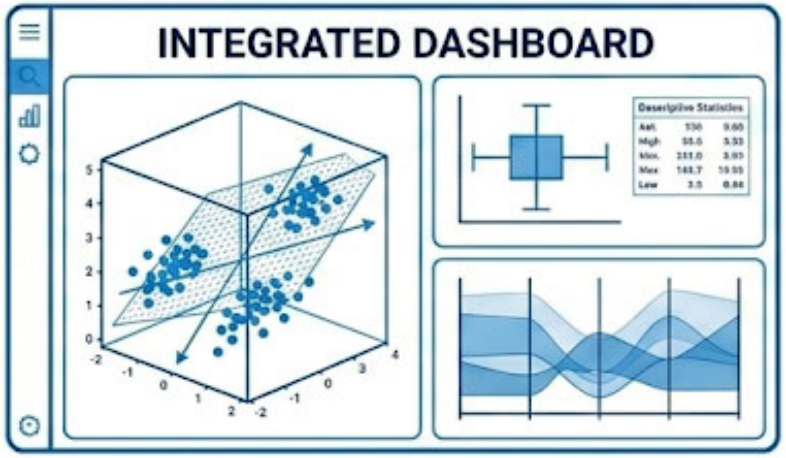


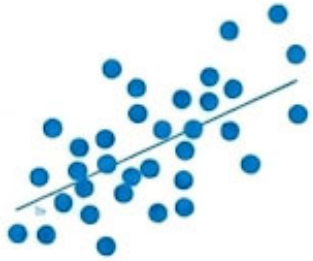
HANDS ON

Step 4 & 5: Data Analysis & Visualization

(Focus: Applied Statistics & Advanced Visuals)







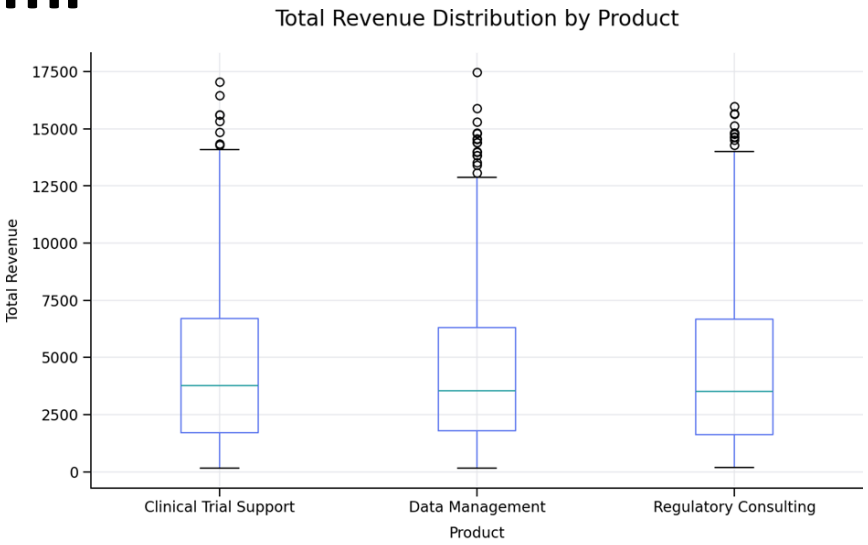
HANDS ON - Statistical Model & Forecasting

[Task 1] Revenue Distribution Analysis by Product Type
Analyze how Total_Revenue differs across:
Data Management
Regulatory Consulting
Clinical Trial Support

Methods
Descriptive statistics:
- Mean, median, standard deviation of Total_Revenue
Visualization:
- Boxplots or histograms

Task 1 – Result Generated by Dr. Aimi

Product	Mean	Median	Standard Deviation
Clinical Trial Support	4,653.96	3,774.19	3,504.52
Data Management	4,575.54	3,544.72	3,505.44
Regulatory Consulting	4,560.62	3,522.26	3,521.73



🔍 Interpretation

- Clinical Trial Support has the highest mean and median revenue, indicating slightly stronger revenue performance.
- Median values are consistently lower than means, suggesting right-skewed distributions (presence of high-value outliers).
- Standard deviation is high and similar across all products, indicating:
  - Large variation in invoice sizes
  - Comparable revenue volatility across services

🔍 Boxplot Insights

- All three products show:
  - Wide interquartile ranges
  - Multiple high-value outliers
- Clinical Trial Support exhibits:
  - Slightly higher upper quartile
  - More extreme upper outliers
- Revenue distributions across products are similar in shape, indicating comparable pricing structures with different deal sizes.

Task 1 – Result Generated by Dr. Dwi

Product Category	Records	Total Revenue	Average Revenue	Median Revenue	Std. Dev.
Clinical Trial Support	532	2,475,907.57	4,653.96	3,774.19	3,504.52
Data Management	534	2,443,338.05	4,575.54	3,544.72	3,505.44
Regulatory Consulting	533	2,431,503.48	4,561.92	3,513.51	3,524.91

Key Findings

1. Clinical Trial Support

- Generates the **highest total revenue**: **2.476 million**.
- Has the **highest average revenue per transaction**: **4,653.96**.
- Also has the **highest median revenue**: **3,774.19**.
- This suggests Clinical Trial Support tends to produce slightly larger transactions than the other services.

2. Data Management

- Total revenue: **2.443 million**.
- Average revenue: **4,575.54**.
- Performs slightly below Clinical Trial Support but above Regulatory Consulting.
- Revenue distribution is very similar to the other categories.

3. Regulatory Consulting

- Total revenue: **2.432 million**.
- Average revenue: **4,561.92**.
- Has the lowest average and total revenue among the three categories, though the difference is relatively small.

Revenue Ranking

Based on **total revenue**:

1. **Clinical Trial Support** → 2,475,907.57
2. **Data Management** → 2,443,338.05
3. **Regulatory Consulting** → 2,431,503.48

Based on **average revenue per transaction**:

1. **Clinical Trial Support** → 4,653.96
2. **Data Management** → 4,575.54
3. **Regulatory Consulting** → 4,561.92

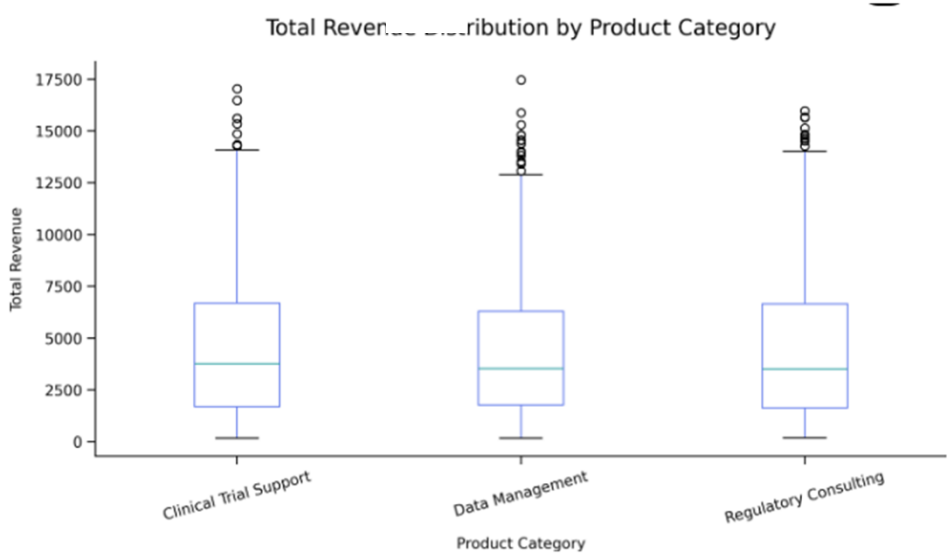
Conclusion

The three product categories perform very similarly, but **Clinical Trial Support** is the strongest revenue contributor, generating about:

- **1.33% more total revenue than Data Management**
- **1.83% more total revenue than Regulatory Consulting**

The standard deviations are nearly identical (~3,500), indicating that revenue variability is similar across all three services. The main difference is that Clinical Trial Support consistently achieves slightly higher transaction values.

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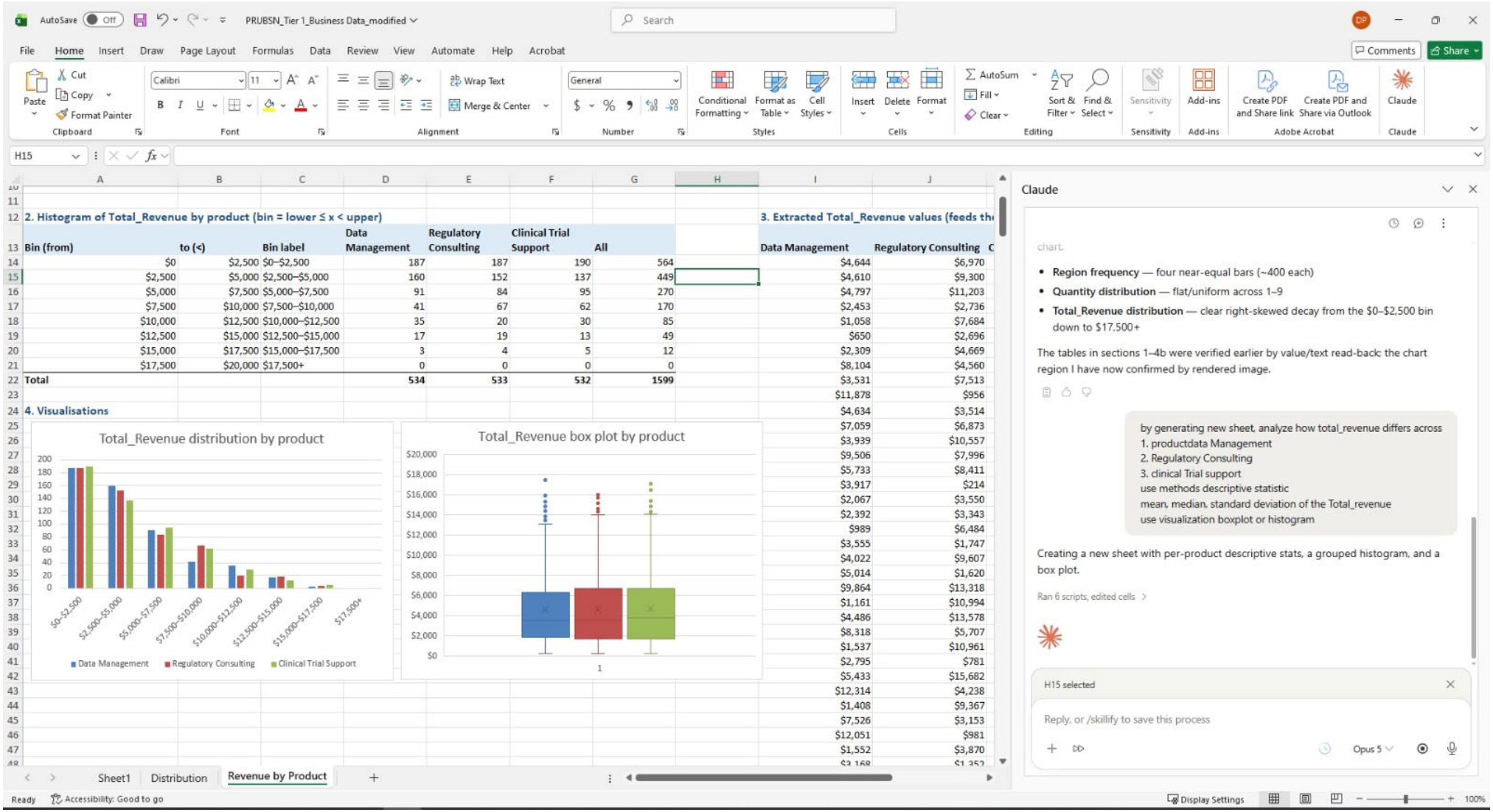


Interpretation

- **Clinical Trial Support** has the highest median revenue (center line of the box), consistent with the earlier statistical summary.
- The **interquartile ranges (IQRs)** are very similar across all three categories, indicating comparable revenue dispersion.
- All three categories exhibit numerous **high-value outliers**, suggesting occasional exceptionally large transactions.
- The overall spread and variability are nearly identical, which aligns with the similar standard deviations (~3,500) observed in the descriptive statistics.

Overall, the boxplot confirms that while **Clinical Trial Support** tends to generate slightly higher revenues, the three product categories have very similar revenue distributions.

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HANDS ON - Statistical Model & Forecasting

[Task 2] Forecastable Trend: Future Total Revenue Trend

Trend to forecast:
Total Revenue over invoice sequence or time proxy.
(If no explicit date column exists, invoice number order can be used as a proxy for time.)

Forecasting approach
Aggregate:
- Total revenue per invoice batch or period
Apply time-series methods:
Moving average
Exponential smoothing
Linear trend projection

Visualization:
Actual aggregated revenue
Moving average
Exponential smoothing
Linear trend projection

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PROMPT

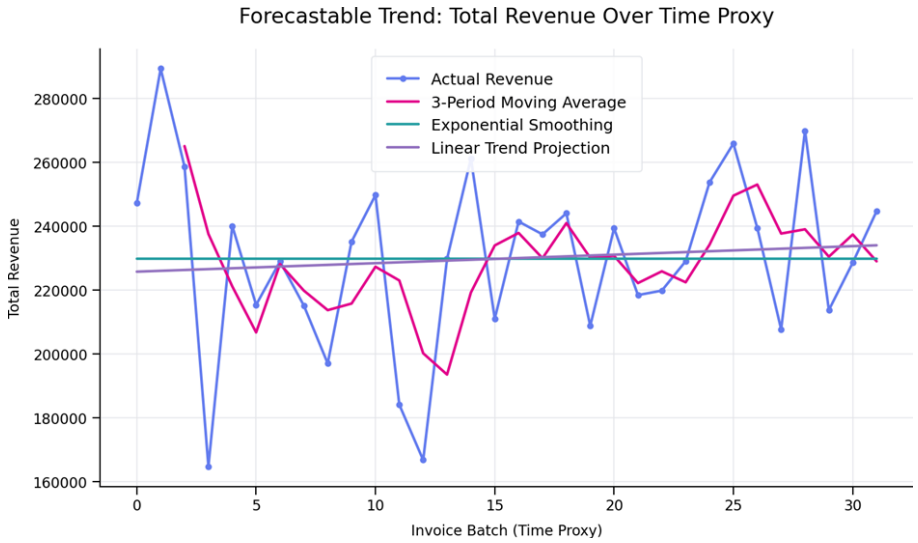
now generate new sheet, forecast trend. trend to forecast: total revenue over invoice sequence or time proxy (if no explicit date column exists, invoice number order can be used as a proxy of time) forecasting

approach aggregate : total revenue per invoice batch or period apply time series method 1) moving average 2) exponential smoothing 3) linear trend projection

visualization actual aggregated revenue moving average exponential smoothing linear trend projection

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Task 2 – Result from Dr Aimi



✅ Interpretation of Results

🔍 Key Observations

Revenue is generally stable with mild growth

Outliers exist, but aggregation reduces their impact

No evidence of sustained revenue decline

Linear trend suggests incremental long-term growth

🔮 What Can Be Forecasted

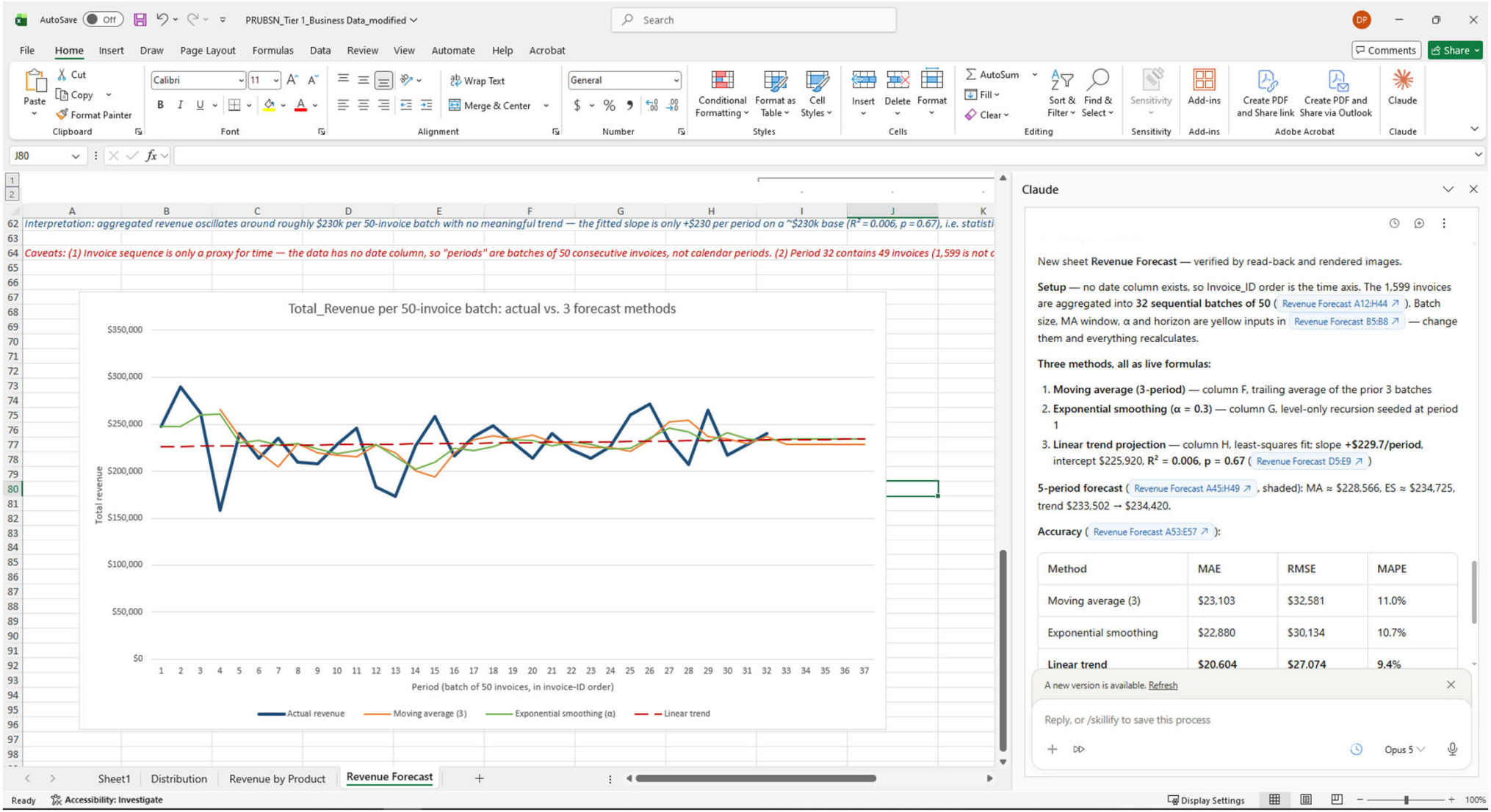
Near-term revenue using exponential smoothing

Medium-term revenue trajectory using linear trend

Expected revenue per upcoming invoice batches

✅ Conclusion (Sample)

Using invoice sequence as a time proxy, total revenue exhibits a stable pattern with gradual positive growth. Moving average and exponential smoothing models indicate short-term stability, while linear trend projection suggests modest long-term revenue expansion. These findings support the use of trend-based forecasting for budgeting and sales planning in the absence of explicit temporal data.



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Make your data work for you, not the other way around.

~~Manual-labor more.~~

Analyze less.

Analyze more.

Manual-labor less.

Module 3 Complete